

Experiment 7

Problem Statement:

Develop a dashboard and reporting tool based on real-time social media data. Develop a real time dashboard using Streamlit, Dash, or PowerBI. Incorporate dynamic reporting with live data from APIs.

Theory:

What is Business Intelligence?

Business intelligence is the process of analyzing business data and presenting it in user-friendly graphical ways such as dashboards, reports, charts, graphs, etc. These reports help the business owners to drive changes, eliminate inefficiencies, and adapt to the market changes. There are many business intelligence (BI) tools that help in building business intelligence reports.

Nearly 50% of companies use BI tools, and many companies are opting to integrate BI tools into their system. Tools like Microsoft Power BI are used by over 3000 companies for business intelligence. Let's discuss more about Power BI.

What is Power BI?

Power BI is the collection of software apps, services, and connectors that turn the raw data into meaningful, immersive, and interactive insights.

It was developed by Microsoft and is very versatile to use. It can accept input from Excel sheets, databases, or even cloud storage. You can connect the data sources, derive insights, and share them with others. Power BI is the most popular business intelligence tool out there, but why? Let's discuss why is Power BI in demand and what makes it different from other tools.

Why Power BI?

In today's world, "data is the new oil". Organizations need a tool that can help them understand the large amount of data that they are collecting. Microsoft Power BI was introduced to fulfill their needs, It is a powerful data visualization and analysis tool that allows businesses to turn raw data into actionable insights and reports.

Power BI helps users who are beginners who do not come from an analytical or tech background. Still, they can create reports on the Power BI desktop.

Microsoft Power BI comes with a free or paid version. The free version only provides Power BI tools like Power BI Desktop and Power Q&A to dashboards. Whereas, in the Pro version they provide services like live report sharing, Power View, and more Power BI apps.

Now that we are familiar with Power BI, let's see how to use Power BI by installing it on our devices.

Microsoft Power BI Installation

Power BI Desktop is available in 32-bit and 64-bit versions. The Power BI desktop app is used to create reports, while Power BI Services (Software as a Service – SaaS) is used to publish the reports, and the Power BI mobile app is used to view the dashboards and reports.

Overview of the Importance of Real-Time Data Analysis

Real-time data analysis is crucial in today's fast-paced digital world, where timely insights can drive better decision-making. It involves processing and analyzing data as it is generated, allowing businesses and organizations to respond to events instantly. This is particularly valuable in industries like finance, healthcare, marketing, and cybersecurity, where delays in decision-making can lead to missed opportunities or significant risks.

Key benefits of real-time data analysis include:

- Immediate Insights – Helps organizations react quickly to trends, threats, or opportunities.
- Improved Efficiency – Reduces the need for manual data processing, leading to faster decision-making.
- Enhanced Customer Experience – Enables businesses to personalize services based on live data.
- Competitive Advantage – Organizations leveraging real-time analytics can stay ahead of competitors.

The Role of Dashboards in Decision-Making

Dashboards play a vital role in transforming raw data into meaningful, visual insights that assist decision-makers. They provide a centralized platform to monitor, analyze, and interact with key metrics in real-time.

Key advantages of dashboards include:

1. Data Visualization – Converts complex data into easy-to-understand graphs, charts, and tables.
2. Centralized Information – Combines data from multiple sources into a single view.
3. Real-Time Monitoring – Helps track key performance indicators (KPIs) as events unfold.
4. Customization & Interactivity – Users can apply filters, drill down into data, and generate reports tailored to their needs.
5. Better Decision-Making – Enables organizations to make data-driven choices based on current trends and insights.

Why Social Media Data is Valuable for Analytics

Social media platforms generate vast amounts of data every second, making them a goldmine for businesses, marketers, and researchers. Analyzing this data in real-time can help organizations understand customer behavior, market trends, and public sentiment.

Some key reasons why social media data is valuable include:

- Brand Monitoring – Companies can track mentions, reviews, and customer feedback to improve products and services.
- Market Trends & Consumer Behavior – Analyzing engagement, likes, shares, and comments provides insights into what customers prefer.
- Sentiment Analysis – AI-driven tools can determine whether public sentiment toward a brand or topic is positive, negative, or neutral.
- Crisis Management – Detecting negative trends early allows businesses to take corrective actions before issues escalate.
- Competitive Analysis – Companies can track competitors' social media activities and measure their own performance against industry benchmarks.

By integrating real-time social media data into dashboards, businesses can stay ahead of trends, improve engagement strategies, and make informed decisions quickly.

Programming Languages for Real-Time Dashboards and Reporting

When developing a real-time dashboard for social media analytics, choosing the right programming language and tools is crucial. The main options for implementation include Python (for Streamlit and Dash) and Power BI for visualization.

Python for Streamlit and Dash

Python is one of the most widely used programming languages for data analysis and visualization. It provides powerful libraries for fetching, processing, and displaying data dynamically.

1. Streamlit

Streamlit is an open-source Python library designed specifically for building interactive web applications with minimal coding effort. It is particularly useful for data scientists and analysts who want to create dashboards without needing extensive web development skills.

Key Features of Streamlit:

- Simple syntax – Build dashboards quickly with just a few lines of Python code.
- Supports real-time updates – Ideal for displaying live social media data.
- Integration with data science libraries – Works well with Pandas, NumPy, Matplotlib, and Plotly.
- Easy deployment – Can be hosted on cloud platforms like Streamlit Sharing, AWS, or Google Cloud.